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Predicting Mangrove Carbon Stock in Riau Province: A Machine Learning Approach Using Sentinel-2 Satellite Imagery

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***Abstract*—Mangrove forests play a role in preserving biodiversity and capturing carbon, which helps combat climate change. Unfortunately, these vital ecosystems are facing increasing threats from deforestation in areas. This research focuses on developing efficient methods to monitor and safeguard mangrove forests by using satellite images and advanced machine learning techniques and also predicting mangrove carbon stock. Various machine learning models like Random Forest, Gradient Tree Boosting and K Nearest Neighbors were compared for their performance in distinguishing mangrove forests from land types in Riau Province, Indonesia. The objective is to classify each pixel in the imagery as mangrove or non-mangrove and estimate the associated carbon stock for conservation purposes. Among the models tested the Random Forest model showed the accuracy of 99.39% making it the preferred choice for calculating carbon stock in the mangrove areas. Based on this models' predictions the estimated value of carbon stock was approximately IDR 1,393,887,913.11. The research shows how well integrating high resolution satellite imagery with machine learning techniques for mangrove monitoring and carbon stock estimation.**

Keywords—carbon stock prediction, climate change, mangrove monitoring, remote sensing , sentinel satellite

I. Introduction

Climate change is an issue that affects globally on various sector that can make many negative effects, one of them is sufficient production and food supplies [1]. Climate change can be caused by natural influence of the environment itself but many of them is likely due to human behavior like the use of fossil fuel that release greenhouse gases (GHG) such as carbon dioxide, methane, etc. into the environment. In the last decade, this problem has caused many serious problem among the other environmental issues. This climate change issue leads to the increase of temperature of the planet which may start due to industrial revolution in the past [2].

Mangrove is one of the way to mitigate climate change, mangrove can prevent many climate change such as coastal erosion, rise of the sea level, severe storms, hurricanes, and tsunami. Mangrove forests can also store and sequester carbon in large capacity [3]. Because of that Mangrove deforestation can caused some problem in the environment due to its significant role. Indonesia, which once had the biggest mangrove cover in the world, has now lost its title due to crisis and continues to lose mangrove cover area annually. If this continues, greenhouse gases (GHG) can accumulate and cause significant damage to the environment [4].

Therefore, there is a need for efficient mangrove monitoring to overview mangrove distribution due to many deforestation that occurred in the last decade. As explored by Lomeo et al. in [19], efforts have been made to monitor and evaluate the spatial-temporal distribution of mangroves using deep learning techniques. In their study, they utilized a cloud-based approach, employing a training dataset consisting of 480 points for mangrove locations with 4 different types of mangroves, which they then multiplied to enhance the dataset. This method highlights the effectiveness of advanced remote sensing in assessing mangrove ecosystems in Southeast Asia. In contrast, this research utilizes a dataset provided by the Ministry of Environment and Forestry, focusing on the prediction of mangrove carbon stock in Riau Province. By leveraging this comprehensive dataset, this study aim to build upon the insights gained from Lomeo et al. and improve the accuracy of mangrove monitoring through machine learning methodologies. This study seeks to address this gap by leveraging Sentinel-2 satellite imagery and machine learning algorithms to develop a more efficient and accurate method for predicting mangrove carbon stock in Riau, Indonesia. By integrating remote sensing data with predictive analytics, this research aims to provide a comprehensive monitoring approach that can be scaled to assess mangrove conditions across larger areas. By accurately predicting the current distribution of mangroves, this study establishes a baseline that can be used for future assessments. This helps in understanding changes over time, whether due to natural factors or human activities.

II. Literature Review

Mangrove deforestation is one of the causes of climate change [5]. This is because mangrove is able to store and sequester carbon at large capacity, preventing a lot of climatic changes like coastal erosion, sea level rise, severe storm, hurricane, and tsunami [3]. Indonesia is faced with challenges in maintaining the area of its mangroves, especially with increasing deforestation by local residents, as it has one of the largest covers of mangrove in the world. Moreover, during the period from 2000 to 2019, the mangrove area in Riau, Indonesia, had been reduced by 2495.9 ha every year [6].

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Remote sensing has become an invaluable tool for mapping and keeping track of mangrove ecosystems [7]. These methods are popular because they offer detailed, high-resolution data over large areas. Many studies have shown that spectral bands and vegetation indices are really effective for identifying mangroves. For instance, the Normalized Difference Vegetation Index (NDVI) and the Enhanced Vegetation Index (EVI) are often used to tell apart vegetated areas from non-vegetated ones. Research suggests that using multiple spectral bands along with these indices can boost classification accuracy. On the other hand, machine learning algorithms, more precisely Random Forest, have been brilliant in pattern recognition [8]. The basic way by which Random Forest works is: it creates many decision trees and trains an ensemble of them. It then later votes on the final decision. It is rather famous because of its high accuracy, avoidance of overfitting, and the ability to work on large datasets with a high dimension of features. Random Forest has been applied successfully to various tasks in the remote sensing field, such as mangrove mapping [9]. This study does not compare results with decision trees. The justification lies in Random Forest's ensemble approach, which typically yields better performance in complex datasets by reducing variance and improving robustness. This allows for a more nuanced analysis of mangrove ecosystems compared to single decision tree classifiers. Spectral bands are capturing specific light wavelengths, and due to this, it makes different bands respond according to their type against various land covers, making them helpful for classification. Vegetation indices are computed from these spectral bands, which help outline certain properties of vegetation. Other standard indices, like NDVI, EVI, and SAVI, increase the contrast between vegetation and other land use/covers and, therefore, optimize their classification accuracy [10].

K-Nearest Neighbors is one of the well-known algorithms for classification. The way it works is to compare a given sample with the nearest neighbors in feature space and decide upon the class based on the majority vote from these neighbors. The KNN belongs to one of the main techniques of remote sensing due to its simplicity and ability to be implemented in complex spatial patterns. It depends crucially on the choice of 'k' (number of neighbors) and the distance metric [11]. Gradient Tree Boosting is a powerful predictive model based on an ensemble of decision trees. While building the trees, the next trees are aimed at correcting the mistakes of the previous ones. Such an approach is very effective in complex data and is adaptive during learning. It has also been applied to a large extent in remote sensing over challenging classification tasks, particularly for large-scale datasets.

Mangroves are effective carbon sinks of both their biomass and soil. Assessment of carbon stock in mangroves is important and has to be understood in a probable way to account for their role in climate change mitigation [12]. Efforts to monitor and evaluate the spatial-temporal distribution of mangroves using deep learning techniques have been highlighted by Lomeo et al. in [19]. Their research employed a cloud-based approach and utilized a training dataset comprising 480 points representing four different types of mangrove species, which they augmented to enhance the dataset. This methodology demonstrates the potential of advanced remote sensing technologies in assessing mangrove ecosystems throughout Southeast Asia. In contrast, the present study leverages a dataset provided by the Ministry of Environment and Forestry, concentrating on predicting the carbon stock of mangroves in Riau Province. By utilizing this extensive dataset, this research aims to build on the findings of Lomeo et al. and enhance the accuracy of mangrove monitoring through various machine learning techniques. This study addresses existing gaps in the literature by integrating Sentinel-2 satellite imagery with machine learning algorithms, thereby developing a more efficient and precise method for predicting mangrove carbon stocks in Riau, Indonesia. The integration of remote sensing data with predictive analytics is intended to offer a comprehensive monitoring framework that can be scaled to evaluate mangrove conditions over larger regions.

III. Study Area

In conducting this study, mangrove ecosystems in Riau region of Indonesia were taken as the main subject and the time frame was from January 1, 2021 to December 31, 2021. Figure 1 illustrates the locations of the eight selected sites, providing a consolidated coverage of mangrove habitats in the area, ranging from the estuary to riverbanks. These included Dumai, Bengkalis, Rokan Hilir, Kepulauan Meranti, Pelalawan, Siak and Indragiri with both conservation and non-conservation areas included. Some of these sites are of important ecological significance and have been recognized by the government which made them popular tourist destination points because of their rich biodiversity.

IV. Methodology

Details of the comparative evaluation of machine learning models for mangrove classification and carbon stock prediction using Sentinel-2 satellite imagery are presented schematically in Fig. 2.

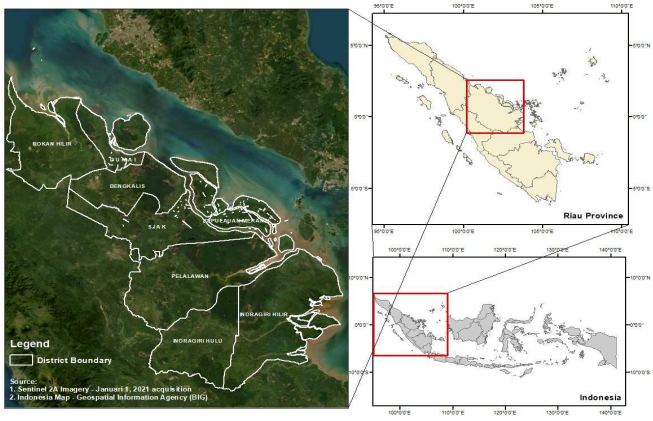

Fig 1. Study Area in Sumatra Island

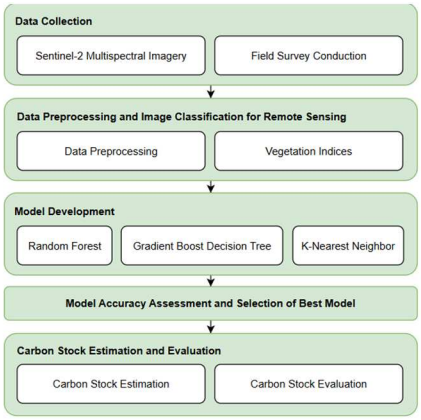

Fig 2. Overview of the Proposed Methodology Workflow

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A. Data Collection

The study utilized a Level 2A Sentinel-2A satellite image acquired throughout 2021 by Ministry of Environment and Forestry as the mangrove dataset [6] and for the other classes, this research used ESA WorldCover 10m 2021 v200 [13]. Detailed specifications of the Sentinel-2A data are presented in Table 1. Image pre-processing and spectral transformations were performed using specialized image processing software.

B. Field Sample Collection

Field data were collected from January 1 to December 31, 2021, using GPS, tally sheets, cameras, and phi-bands for ecological data collection. The SNI 7717 standards were employed to create mapped classes of mangrove density for that period as modified in 2020 to reduce five categories into three namely: dense mangrove, moderate mangrove and sparse mangrove which would enable accuracy enhancement and good representation. Accordingly, this led to mapping out areas where there are different density ranges of mangroves within just a single map rather than having many maps with varying densities. The current condition on the national mangrove map (Peta Mangrove Nasional or PMN) reveals present day status concerning Indonesian forests comprising mainly of mangrove trees based on remote sensing methods observed from space satellites compared to other approaches hence satisfying all specifications needed by this standardization criteria.

The analysis used a Sentinel-2 L2A composite image to create an annual cloud-free composite for the same period. A total of 1,200 field points were sampled, with 70% (840 points) used for model training and 30% (360 points) for testing. Field points were randomly selected based on accessibility. Each point was represented by a 10-by-10-meter square plot, matching the 10-meter resolution of Sentinel-2 images. Transect lines, originally proposed by Kauffman & Donato (2012) as circular plots, were adapted to square plots to align with the pixel shape in the satellite images, ensuring consistency between field and satellite data. Figure 2 illustrates the schematic of the mangrove sampling plot, showing the adaptation of transect lines to align with the Sentinel-2 imagery resolution.

C. Data Processing and Image Classification for Remote Sensing

Using Google Earth Engine (GEE) platform, classification into mangrove and non-mangrove areas was done in this research. In particular, they used images that had less than 30% cloud cover for clarity assured while using

TABLE I. SATELLITE SPECIFICATION

Satellite	Band	Description	Spatial Resolution (m)	Wavelength (nm)
Sentinel 2-L2A	B01	Coastal Aerosol	60	443
	B02	Blue	10	490
	B03	Green	10	560
	B04	Red	10	665
	B05	Vegetation Red Edge	20	704
	B06	Vegetation Red Edge	20	740
	B07	Vegetation Red Edge	20	783
	B08	Near-Infrared	10	842
	B09	Narrow Near-Infrared	20	865
	B10	Water Vapor	60	945
	B11	Short Wave Infrared 16	20	1.613
	B12	Short Wave Infrared 22	20	2.202

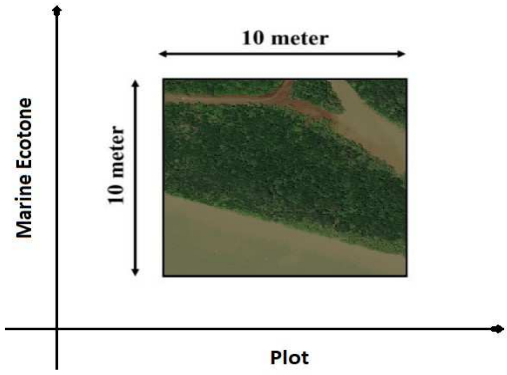

Fig. 3. Schematic of Mangrove Sampling Plot

Sentinel 2L2A dataset. Over one year, this research produced a cloud free composite for the Riau region and its environs. To clear away cloud-affected pixels from the individual images cloud masking was applied which used Quality Assessment (QA) band hence filtering out foggy pixels successfully from each photo. Additionally, median reduction function was used together with median pixel value selection from the set of cloud affected pictures so as to ensure no significantly bright pixels (clouds) and no excessively dark ones (shadows) remained in final image.

For distinguishing between mangrove and non-mangrove areas, this research utilized various classification methods, including Random Forest (RF), K-Nearest Neighbors (KNN), and Gradient Tree Boost (GTB). These models were selected for their strong performance in classification tasks and their ability to handle noisy data effectively. Each method brings unique strengths, with RF combining predictions from multiple decision trees, KNN classifying based on proximity to known data points, and Gradient Tree Boost iteratively builds decision trees, with each new tree focusing on correcting the errors of the previous ones. This method improves predictive accuracy by minimizing the loss function through gradient descent, making it particularly effective for capturing complex patterns in the data. To aid in this process, this research conducted a visual inspection of false-color composites using shortwave infrared (SWIR), near-infrared (NIR), and red bands, which helped in differentiating between non-mangrove areas and mangrove areas more clearly.

Initially, the interactive map feature of Google Earth Engine (GEE) was used to identify specific points (latitude/longitude pairs) for each of the nine Land Cover

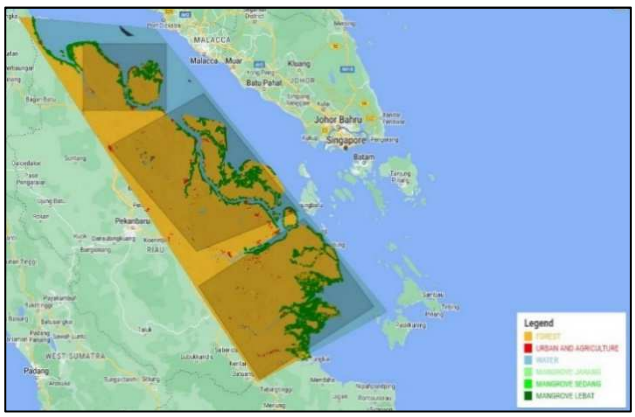

Fig 4. Image of Riau Generated in GEE using Cloud-Masked Sentinel-2 Imagery Captured in 2021

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Types (LCTs). These spots served as the baseline for the creation of mangrove shapefiles and were chosen based on the median image for 2021, in accordance with the period utilized in Worthington et al. [14]. Figure 4 illustrates the Riau region generated in GEE using cloud-masked Sentinel-2 imagery captured in 2021. Considering the different lighting conditions and the heterogeneous character of mangrove habitats, an average of 120 points per LCT were selected from the Riau region of interest (ROI). A total of 1.200 points were obtained from this selection (see Table 2). Despite the large area covered by these locations, they were all located close to the equator, which ensured regular variations in the Sentinel-2 sensor's field of view. Buffers of 5 meters for clouds and 50 meters for other LCTs were applied to collect additional pixels for classification purposes. After an initial exploratory phase, the classification categories were reduced from nine to six by merging agriculture, open fields, and urban areas into a single category.

TABEL II. THE NUMBER OF POINTS FOR EACH CLASS WAS RECORDED IN THE RIAU ROI

	Non Mangrove Classes			Mangrove Classes		
	Forest	Urban and Agriculture	Water	Sparse Mangrove Density	Moderate Mangrove Density	Dense Mangrove Density
Riau	120	120	120	120	120	120

In the next stage, the median composite image was segmented using straightforward, non-iterative clustering techniques. This approach improved classification accuracy by enhancing the differentiation between various spectral regions. Subsequently, a conventional Random Forest (RF) classifier was employed to process the segmented multi-band image. this research evaluate using different models such as Random Forest, K-Nearest Neighbors (KNN), and Gradient Tree Boost (GTB). The classification utilized 21 bands, which included 9 spectral indices and 12 spectral bands (B1–B12) (see Table 3). The classification results were appended to the median composite image as an additional "classification" band, resulting in a total of 22 bands.

TABEL III. SPECTRAL INDICES APPLIED TO THE MEDIAN IMAGE AS BANDS

Vegetation Indices	Equation
NDVI (Normalized Difference Vegetation Index)	$\mathrm{NDVI} = \frac{\mathrm{NIR} - \mathrm{Red}}{\mathrm{NIR} + \mathrm{Red}}$
DVI (Difference Vegetation Index)	$\mathrm{DVI} = \mathrm{NIR} - \mathrm{Red}$
GOSAVI (Green Optimized Soil Adjusted Vegetation Index)	$\mathrm{GOSAVI} = \frac{\mathrm{NIR} - \mathrm{Green}}{\mathrm{NIR} + \mathrm{Green} + 0.16}$
SAVI (Soil Adjusted Vegetation Index)	$\mathrm{SAVI} = \frac{\mathrm{NIR} - \mathrm{Red}}{\mathrm{NIR} + \mathrm{Red} + L^a} \times (1 + L^a)$
NDWI (Normalized Difference Water Index)	$\mathrm{NDWI} = \frac{\mathrm{Green} - \mathrm{NIR}}{\mathrm{Green} + \mathrm{NIR}}$
MNDWI (Modified Normalized Difference Water Index)	$\mathrm{MNDWI} = \frac{\mathrm{Green} - \mathrm{SWIR}}{\mathrm{Green} + \mathrm{SWIR}}$
NDSI (Normalized Difference Snow Index)	$\mathrm{NDSI} = \frac{\mathrm{Green} - \mathrm{SWIR}}{\mathrm{Green} + \mathrm{SWIR}}$
NDMI (Normalized Difference Moisture Index)	$\mathrm{NDWI} = \frac{\mathrm{NIR} - \mathrm{SWIR}}{\mathrm{NIR} + \mathrm{SWIR}}$
EVI (Enhanced Vegetation Index)	$\mathrm{EVI} = \frac{\mathrm{NIR} - \mathrm{Red}}{\mathrm{NIR} + C_1^c * \mathrm{Red} - C_2^c * \mathrm{Blue} + 1} \times G^b$
EVI2 (Two-band Enhanced Vegetation Index)	$\mathrm{EVI2} = \frac{\mathrm{NIR} - \mathrm{Red}}{\mathrm{NIR} + 2.4 \times \mathrm{Red} + 1} \times G^b$

[a.] Soil Brightness Correction, defined as 0.5.
[b.] Gain Factor, defined as 2.5.
[c.] Coefficients of Atmospheric Resistance.

D. Random Forest Model Development

In this research, RF was used for classification since it is efficient in classification problems and feature importance evaluation. RF constructs multiple decision trees from random subsets of data and features, then aggregates their predictions for improved accuracy and robustness. The current study used 100 decision trees for random forest classification. Typically, a number of trees between 64 and 128 is enough to provide high performance [15], especially when the used data set is small or the treated task is simple. Figure 5 illustrates the Random Forest classifier structure.

E. Gradient Boost Decision Tree Model Development

Gradient Boost Decision Tree (GBDT) was chosen for its strength in complex classification tasks by modeling intricate relationships in the data using a sequence of trees. In GBDT, trees are created that correct errors of the previous ones, improving accuracy via gradient descent. In the case of this research, the model was run with 100 trees, and in addition, some important hyper-parameters such as tree depth and learning rate were tuned in order to perform better on this nonlinear data like satellite image processing. Figure 6 illustrates how combining multiple trees enhances predictive accuracy.

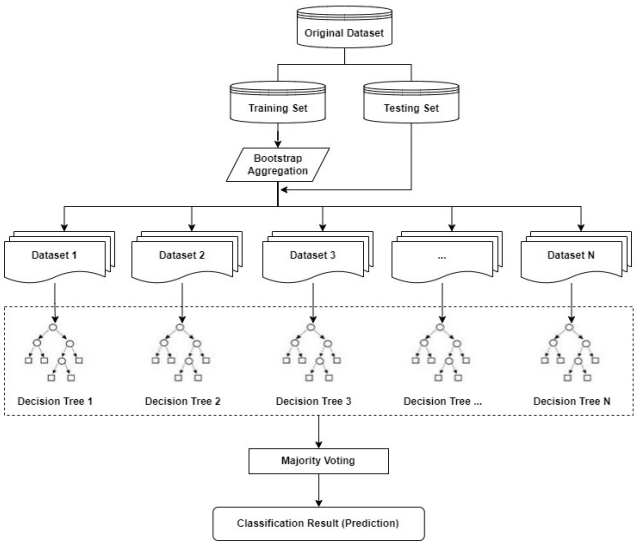

Fig. 5. Random Forest Classifier with Multiple Decision Trees

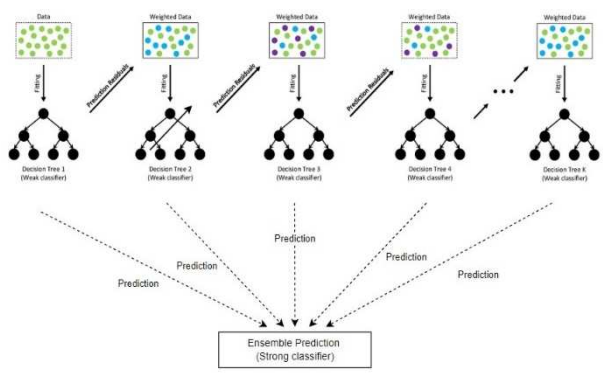

Fig. 6. Gradient Boost Decision Tree Model with Sequential Decision Trees

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F. K-Nearest Neighbors Model Development

The K-Nearest Neighbors (KNN) algorithm was chosen for its simplicity and effectiveness in classification tasks. KNN classifies data points based on their proximity to the 'k' closest training examples, providing insights into classification outcomes by analyzing the majority class among these neighbors. It doesn't require a separate training phase; instead, it computes distances (e.g., Euclidean) between the input and data points in the dataset [16]. In this study, k = 10 was used, offering a balance between bias and variance. Typically, k values between 5 and 15 are selected for robust performance with moderate complexity datasets.

G. Model Accuracy Assessment

The Random Forest (RF) classifier, along with the Gradient Tree Boost (GTB) and K-Nearest Neighbors (KNN) models, were evaluated in Google Earth Engine (GEE) was evaluated based on the Kappa coefficient and a confusion matrix that comprised producer and user accuracies. A 70–30 split between training and testing was used for this assessment. User accuracy indicates the likelihood that the categorized samples are accurate, while producer accuracy gauges how well the categorization fits the real ground truth categories. The classification's overall accuracy is assessed using the Kappa coefficient. The values of each of these indicators range from 0 to 1. Equation (1) provides the Kappa coefficient, which measures the agreement between observed and predicted classifications, taking into account the agreement occurring by chance. The Kappa coefficient is given by:

$$\kappa = \frac{p_o - p_e}{1 - p_e} \tag{1}$$

where p_o is the observed agreement (accuracy), and p_e is the expected agreement by chance. Equation (2) defines the observed agreement. The observed agreement p_o is the proportion of correctly classified instances:

$$p_o = \frac{\sum_{i=1}^{k} n_{ii}}{N} \tag{2}$$

where n_{ii} is the number of correctly classified instances in class i, k is the number of classes, and N is the total number of instances. The expected agreement pe is calculated based on the confusion matrix's marginal totals. Equation (3) defines the expected agreement:

$$p_e = \sum_{i=1}^{k} \left(\frac{n_{i+} \times n_{+i}}{N^2} \right) \tag{3}$$

where n_{i+} is the total number of instances in the true class i, and n_{+i} is the total number of instances classified as class i. The Kappa coefficient ranges from -1 to 1. Table 4 defines the interpretation of the Kappa coefficient and its values:

TABEL IV. INTERPRETATION OF KAPPA COEFFICIENT VALUES

Kappa Value	Interpretation
$\kappa \leq 0$	No agreement (or agreement worse than chance)
$0 < \kappa \leq 0.20$	Slight agreement
$0.21 \leq \kappa \leq 0.40$	Fair agreement
$0.41 \leq \kappa \leq 0.60$	Moderate agreement
$0.61 \leq \kappa \leq 0.80$	Substantial agreement
$0.81 \leq \kappa \leq 1$	Almost perfect agreement

In addition to the Kappa coefficient, user accuracy and producer accuracy provide important insights into the performance of the classification. Equation (4) provides the user accuracy, also known as commission error, reflects the probability that a classified instance truly belongs to the class it was assigned to. It is calculated by:

$$\mathrm{UA}_i = \frac{n_{ii}}{n_{+i}} \tag{4}$$

Here, n_{ii} represents the number of correctly classified instances for class i, while n_{+i} is the total number of instances classified as class i. User accuracy thus indicates the reliability of the classifier when it assigns an instance to a particular class.

On the other hand, Equation (5) provides the producer accuracy, also known as omission error, which measures the probability that an instance of the true class is correctly classified. It is given by:

$$\mathrm{PA}_i = \frac{n_{ii}}{n_{i+}} \tag{5}$$

In this case, n_{ii} is the number of instances correctly classified in class i, and n_{i+} is the total number of instances in the true class i. Producer accuracy reflects how well the classifier captures all instances of each true class.

V. CARBON STOCK ESTIMATION AND VALUATION

Carbon sequestration plays a critical role in mitigating climate change by capturing and storing atmospheric carbon dioxide (CO_2). This process is especially important in ecosystems such as mangroves, forests, and wetlands, which act as significant carbon sinks. Estimating the carbon sequestration value involves quantifying the amount of carbon stored within these ecosystems and converting it into (CO_2) equivalents to assess its impact and economic value. Understanding this value helps in the evaluation of ecosystem services and supports efforts in conservation and carbon credit markets. Equation (6) defines the calculation of the estimation of carbon sequestration value:

$$\mathrm{eCO_2} = \frac{M_{\mathrm{CO_2}}}{A_c} \times \text{Stock C} \tag{6}$$

In this equation, $M_{\mathrm{CO_2}}$ represents the relative molecular weight of $\mathrm{CO_2}$ (44), A_c is the relative atomic weight of carbon (12), and Stock C refers to the carbon stock in tons per hectare (tons/ha). For instance, research conducted in the Riau Islands has reported an average total carbon stock of 2052.78 tons C/ha [17], highlighting the significant carbon storage capacity of mangrove ecosystems in this region.

Furthermore, Equation (7) defines the monetary valuation of this carbon stock, termed Carbon Pool Valuation (CPV), is calculated to assess the economic value of the carbon stored in mangrove forests. This valuation is conducted using the following formula:

$$CPV = \sum_{i=1}^{m} \left(A(EF) \left(\frac{44}{12} \right) (\$5) \right) \tag{7}$$

where A represents the mangrove land cover class as predicted by our model, and EF is the Emission Factor, which is replaced by the average carbon stock value for all known

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mangrove species in Riau, measured at approximately 2052.78 tons C/ha. The factor is used to convert carbon (C) to its CO2 equivalent, and the value of $5 represents the carbon price per tCO2e, based on the opening and closing prices on the Indonesia Carbon Exchange (IDXCarbon) during its launch on 26 September 2023 [18].

RESULT AND ANALYSIS

In this study, this research evaluated three machine learning models—Random Forest (RF), Gradient Tree Boost (GTB), and K-Nearest Neighbors (KNN)—for mangrove classification and carbon stock prediction using Sentinel-2 satellite imagery. The performance metrics for each model, including the overall accuracy, kappa coefficient, producer's accuracy, and consumer's accuracy, are shown below.

The RF model achieved the highest test accuracy of 99.36%, with a kappa coefficient of 0.9921, indicating a near-perfect agreement. The producer's accuracy for all classes was above 96%, with most classes achieving perfect accuracy. The confusion matrix (Table V) shows minimal misclassification, particularly in the mangrove and water classes.

The Gradient Tree Boost model performed comparably to the RF model, with a test accuracy of 99.33% and a kappa coefficient of 0.9916. The consumer's accuracy was slightly lower in certain classes compared to RF, indicating a marginally higher rate of misclassification. The confusion matrix (Table VI) illustrates similar patterns of misclassification as observed in the RF model, particularly in the forest and urban classes.

TABEL V. RANDOM FOREST CONFUSION MATRIX RESULT

Random Forest		Truth Data						User's Accuracy
		Sparse Mangrove Density	Moderate Mangrove Density	Dense Mangrove Density	Forest	Urban	Water	
Classifier Result	Sparse Mangrove Density	2751	23	0	0	0	0	99.171%
	Moderate Mangrove Density	0	2661	0	0	18	2	99.254%
	Dense Mangrove Density	65	34	2559	0	0	0	96.275%
	Forest	0	0	0	6606	0	0	100%
	Urban	0	0	0	0	2606	0	100%
	Water	0	0	0	0	0	4962	100%
Producer's Accuracy		97.692%	97.903%	100%	100%	99.314%	99.96%	$\kappa = 0.992$

TABEL VI. GRADIENT TREE BOOST CONFUSION MATRIX RESULT

Gradient Tree Boost		Truth Data						User's Accuracy
		Sparse Mangrove Density	Moderate Mangrove Density	Dense Mangrove Density	Forest	Urban	Water	
Classifier Result	Sparse Mangrove Density	2749	24	0	0	0	1	99.099%
	Moderate Mangrove Density	0	2658	0	0	18	5	99.142%
	Dense Mangrove Density	65	34	2559	0	0	0	96.275%
	Forest	0	0	0	6606	0	0	100%
	Urban	0	0	0	3	2603	0	99.885%
	Water	0	0	0	0	0	4962	100%
Producer's Accuracy		97.69%	97.865%	100%	99.955%	99.313%	99.879%	$\kappa = 0.992$

TABEL VII. K-NEAREST NEIGHBORS CONFUSION MATRIX RESULT

K-Nearest Neighbors		Truth Data						User's Accuracy
		Sparse Mangrove Density	Moderate Mangrove Density	Dense Mangrove Density	Forest	Urban	Water	
Classifier Result	Sparse Mangrove Density	2651	23	94	1	3	2	95.566%
	Moderate Mangrove Density	0	2495	165	0	20	1	93.062%
	Dense Mangrove Density	0	2	2656	0	0	0	99.925%
	Forest	0	1	0	6605	0	0	99.985%
	Urban	0	0	0	0	2606	0	100%
	Water	0	0	0	0	0	4962	100%
Producer's Accuracy		100%	98.969%	91.115%	99.985%	99.125%	99.94%	$\kappa = 0.983$

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The KNN model, while still highly accurate, exhibited the lowest performance among the three models, with a test accuracy of 98.6% and a kappa coefficient of 0.9826. The producer's accuracy for the urban class was significantly lower at 93.06%, indicating that this class was more prone to misclassification in the KNN model. The confusion matrix (Table VII) shows more substantial misclassification, particularly between the urban and forest classes.

The results indicate that while all three models provide high classification accuracy, the Random Forest model slightly outperforms the other two in both overall accuracy and kappa coefficient. The KNN model, although accurate, struggles with the urban class, which could be attributed to the nature of the KNN algorithm's sensitivity to the distribution of data in feature space. The Gradient Tree Boost model, while nearly as accurate as Random Forest, shows a marginally higher rate of misclassification in the confusion matrix.

When examining the total mangrove pixel counts predicted by each model, a clear variation in their performance is evident. The Random Forest model identified a total of 3,703,770 mangrove pixels, corresponding to an area of approximately 37,037.7 hectares and a valuation of approximately IDR 1,393,887,913.11. The K-Nearest Neighbors (KNN) model, on the other hand, predicted a higher total of 4,217,901 mangrove pixels, resulting in an area of about 42,179.01 hectares and a valuation of about IDR 1,587,377,516.04. The Gradient Tree Boost model identified 3,919,835 mangrove pixels, which translates to an area of approximately 39,198.35 hectares and a valuation of about IDR 1,475,202,463.41. These differences in pixel counts and corresponding area valuations reflect the varying sensitivity of each model to the features used in the classification task

CONCLUSION

This research aimed to address the pressing issue of mangrove deforestation by identifying the most suitable machine learning model for accurately classifying mangrove and non-mangrove areas using Sentinel-2 satellite imagery. The primary objective was to select a model that could not only distinguish mangrove regions with high accuracy but also facilitate the estimation of carbon stock values, which are crucial for assessing the ecological and economic value of these areas.

The study evaluated three machine learning models—Random Forest (RF), Gradient Tree Boost (GTB), and K-Nearest Neighbors (KNN). Among them, Random Forest emerged as the best-performing model, achieving a test accuracy of 99.36% and a kappa coefficient of 0.9921. This model accurately identified a mangrove area of approximately 37,037.7 hectares, corresponding to a carbon stock valuation of IDR 1,394,010,137.52. The superior performance of Random Forest indicates its robustness in differentiating mangrove areas from other land cover types, such as water, forest, and urban areas, with minimal misclassification.

While the Gradient Tree Boost model also performed well, with a test accuracy of 99.33% and a kappa coefficient of 0.9916, it showed slightly higher misclassification rates, particularly in the forest and urban classes. The KNN model, though accurate, lagged behind with a test accuracy of 98.6% and a kappa coefficient of 0.9826, struggling notably with the urban class and resulting in overestimated mangrove areas.

The successful identification of mangrove areas enables precise carbon stock calculations, crucial for understanding the environmental impact and economic benefits of mangrove conservation. By leveraging the best-performing model, this research can further extend this approach to assess time series data, allowing us to monitor mangrove deforestation trends over time. This temporal analysis will be essential for evaluating changes in carbon stocks and devising strategic interventions for mangrove conservation and restoration.

The findings underscore the importance of accurate classification in environmental monitoring, as it directly impacts carbon valuation and policy decisions. By utilizing the best model, stakeholders can more effectively track mangrove deforestation, assess carbon stock changes, and make data-driven decisions to mitigate environmental degradation. Future research should focus on integrating these models with time series data to further validate the models' effectiveness in real-time monitoring and carbon stock valuation, providing a comprehensive tool for sustainable mangrove management and conservation efforts.

AUTHOR'S CONTRIBUTION

The study was conceived and designed by Marvin Luckianto, Renaldi Addison, Alexander Agung Santoso Gunawan and Fabian Surya Pramudya. The dataset, provided by the Ministry of Environment and Forestry, was supplied by Anton Dwi Fitriyanto. The writing is guided by Alexander Agung Santoso Gunawan and Fabian Surya Pramudya. All authors read and approved the manuscript.

AVAILABILITY DATA AND MATERIALS

This study utilized publicly available Sentinel-2 Level-2A (L2A) satellite imagery, accessed via Google Earth Engine, to obtain high-resolution spatial information.

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